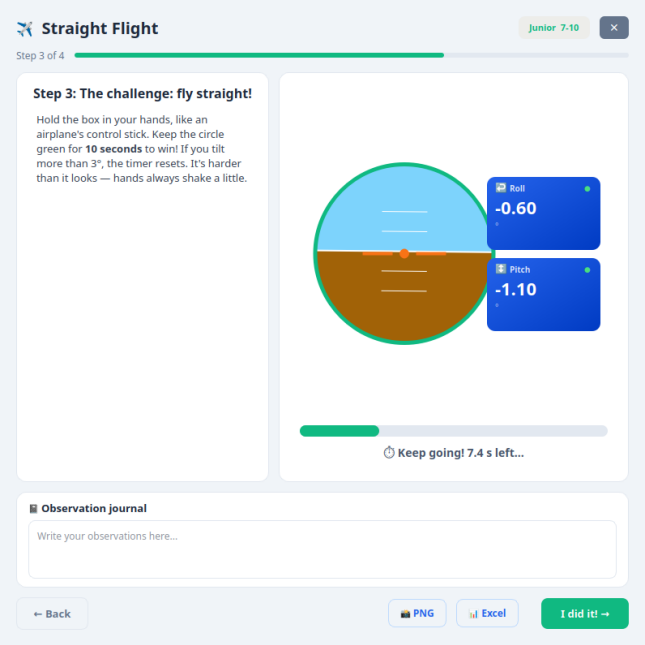
Flying Straight

The box becomes a spirit level and an artificial horizon — and ten seconds of stillness turn out to be surprisingly hard

Level: Junior — ages 7–10 (Romanian school years II–IV) • **Time in class:** 15 min, or one 50-minute lesson • **Sensors:** MPU-6500 (acceleration)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Junior — ages 7–10 (Romanian school years II–IV). The numbers are small and signed — −7°, +2° — but **a child does not need to read them** in order to play: the circle on screen turns green or it does not. The numbers come into the discussion from year III, and their sign can be handled simply, as "one way" and "the other way".

Assumed knowledge

Nothing compulsory. Comparing two small numbers helps, but the challenge can be done watching only the colour of the circle. Angles as a syllabus topic are **not** required.

Time in class

Short: 15 min — the wizard's steps. Full: one 50-minute lesson, with the tour of surfaces in the room, the reversal test and the discussion about pilots' instruments.

Teacher preparation

5 min — find **one surface you know to be level** and one you know is not; and try the challenge yourself once, so that you know how hard it is (it is harder than it looks).

Sensor settling time

None. The accelerometer responds instantly. For the digital level the box *rests on the surface* with hands off it; for the challenge it also starts from a flat surface and is only then lifted, in both hands.

Sensors and boxes	MPU-6500. **The box is mounted upside down and the firmware already corrects for it**: a box lying flat reports $AZ \approx +1$ g. A box reading -1 g flat is running old firmware (reflash it) or has its sensor fitted the other way up (the **Motion sensor upside down** tick in **Settings**). One box is enough for the whole class with the screen projected; ideally, one per 4–6 children.

Three routes

Short	15 min: the wizard's steps — what roll and pitch are, the digital level, the ten-second challenge, and why pilots need instruments.
Full	The 50-minute lesson: the short route plus **the tour of surfaces** — the desk, the window sill, the floor, the top of a cupboard, a book placed underneath —, the 180° reversal test, and ranking the surfaces by how level they are.
Project	Extension: the challenge with eyes closed for whoever holds the box, with a classmate saying only "higher" or "lower" — which is exactly what an instrument does for a pilot who cannot see outside.

1. Where this fits in the curriculum

- **Mathematics and exploring the environment** (years I–II) — orientation in space and an object's positions: tilted one way, tilted the other, level. **The words "horizontal" and "tilted" gain an instrument that checks them**, which is rare at this age.
- **Natural sciences** (years III–IV) — balance and the forces acting on a body; and, more simply, the idea that an instrument can know something the eye cannot: whether a table really is level.
- **Mathematics** (from year III) — comparing small numbers and the idea of a *tolerance*: not "zero" but "smaller than five". This is a very mature idea, met here in a form a nine-year-old can use. The angles are only read as numbers; computing them from the accelerations takes trigonometry, which arrives in year VII, and stays in the teacher's background.
- **Physical education** — fine control of the hand and quiet breathing; ten seconds of stillness is a real test of motor control.
- **Cross-curricular** — why aircraft have instruments, and what happens when a pilot enters cloud. **This is the question that stays with children**, and the app's last step asks it.

The core skill is **trusting an instrument when the senses mislead**. A child who holds the box "level" and sees 5° on screen has, for the first time, proof that their own judgement is not enough — and that is exactly a pilot's situation in cloud.

2. Time plan

Route	Time	What happens
Short	**15 min**	What roll and pitch are (3), the written prediction (2), the digital level with its two readings (3), **Table or box?** (1), the ten-second challenge (4), why pilots need instruments (2). These are exactly the wizard's six steps.
Full	**50 min**	The short route unhurried (15), the tour of surfaces with five or six marked places and the 180° reversal test (18), the challenge for every child in turn (10), the debrief (7).
Extension	+10 min	The "in cloud" challenge: whoever holds the box closes their eyes, and a classmate says only "higher" or "lower".

Ten seconds are harder than they look — and that is the point

The challenge requires **both roll and pitch** to stay below **five degrees** for **ten seconds**, and the timer *resets* at the first departure. Try it yourself before the lesson: most adults need several attempts.

How it goes, in the app's order. The box is first set down *on a flat surface* — the circle turns green by itself, and the child sees what the target looks like. Then they lift it **in both hands**, like an aircraft's control stick, and try to keep the circle green. **Three things make it possible**, and they are worth saying to the children at the start: both hands, not one; elbows resting on the table; and quiet breathing. The five-degree tolerance is chosen so that a box held like that can be won after a few attempts.

Failing the first attempts is the good part. A child who succeeds first time has learned nothing; one who succeeds on the fourth has discovered in the meantime what needed changing — and that is called fine adjustment.

3. Materials

Item	Per group	Notes
MagicBox with MPU-6500	1	One box and the projected screen works very well for a whole class; ideally, one per 4–6 children.
Laptop with the app	1	USB or WiFi. **On a cable, the cable pulls the box** — leave it slack or work over WiFi.
Surfaces to test	4–6	The desk, the sill, the floor, the top of a cupboard, a shelf. Marked with labels so that every group tests the same ones.
A book or two	1–2	Placed under one edge to make a deliberately tilted surface — the activity's control case.
A spirit level	1 (optional)	If the school has one, comparing with it is very convincing: two independent instruments saying the same thing.
The class clock	1	The app times the challenge itself, but a visible clock helps at the debrief.

Preparation. Find in advance **one surface that is level** and **one that is not** — window sills usually have a slight slope, put there deliberately so that water runs off, and that is an excellent discovery. Mark the places with labels. And try the challenge once, so that you know what to expect.

4. Scientific background for the teacher

How a box with no windows knows it is tilted

The accelerometer does not sense tilt. It senses **acceleration**, and at rest the only acceleration present is gravity: a vector pointing always downwards, of magnitude 1 g.

When the box is level, all of that vector falls on the vertical axis: $a_z = 1$ g, and a_x and a_y are zero. When the box tilts, the vector stays where it is — gravity does not tilt — but *the box's axes* rotate relative to it, so part of it appears on the horizontal axes. From the ratio

of those components the angles are recovered:

$$\text{roll} = \arctan\frac{a_y}{\sqrt{a_x^2 + a_z^2}}, \qquad \text{pitch} = \arctan\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}.$$

The formulas are for the teacher: trigonometric functions arrive only in year VII, and here the children use nothing but the numbers on screen.
This is exactly the idea of a spirit level, only with the bubble replaced by a measurement. And it is the idea of an aircraft's *artificial horizon* — the instrument that shows the pilot where the horizon is when they cannot see it.
Two notes for the teacher. First, the method works **only at rest**: if you move the box, the acceleration of the movement adds to gravity and the angles become false — which is why a real aircraft also uses gyroscopes. Second, in the formulas above a_z **enters squared**, so its sign changes neither the size of the angles nor the green of the circle. A box with its sensor the other way up (-1 g flat) would still show pitch with its sign reversed on the horizon; the remedy lies outside this activity — see the technical note.

4.1 The reversal test: the surface or the box?

No sensor reads exactly zero on a perfectly level surface: every box has a small **zero offset** of its own, usually under one degree. On screen, the offset and the tilt of the surface add up and cannot be told apart from a single reading. **The reversal test separates them**, and it is the same test carpenters use to check a spirit level.

Set the box on the surface and note roll and pitch. Then **turn it round by** 180° **on the spot** — the other end towards the child, in exactly the same place — and read again. The tilt of the surface *changes sign* when the box is turned round; the box's offset does not. So:

- if the number changes sign and stays about as large (+3°, then −3°), **the surface is tilted**;
- if it stays the same, with the same sign (+0.6° both times), **it is the box's offset**, and the surface is level;
- in general, **half the difference** of the two readings is the tilt of the surface and their **average** is the box's offset.

For the younger years the first phrasing is enough: *if the number changes sign when I turn the box round, the table is to blame; if not, the box is.* The half-the-difference calculation is for the year IV children who ask for it.

4.2 Why five degrees, and why ten seconds

Five degrees is a tolerance chosen for a box held in the hands: tight enough that it cannot be hit by accident, wide enough that an eight-year-old can reach it after a few attempts. At three degrees, as it used to be, hand tremor and breathing threw the circle out of green more often than anybody could hold it. It is still worth showing how little it is: over the 50 cm length of a desk, five degrees means one end raised by about 4.4 cm — about the width of three fingers.

Ten seconds is far longer than it sounds. The human hand has a natural tremor, and breathing moves the whole body; that is why both hands and elbows resting on the table help. **The timer resetting at the first departure** is what turns the challenge from luck into a skill: it is not enough to hit the target, you have to *stay* there.

4.3 Why pilots become lost in cloud

The app's last step touches a real and serious subject. With no view outside, the human inner ear **cannot distinguish** a steady bank from level flight: after a few tens of seconds of a gentle turn, the sensation of banking fades, and the pilot "feels" level while actually turning.

The phenomenon is called spatial disorientation and it has caused well-known accidents. The instrument — the artificial horizon — is the only thing that contradicts the pilot, and instrument flight training is largely a matter of learning **to believe the instrument against your own sensation**.

For year IV, the right phrasing is simple and true: *sometimes your body lies to you, and the instrument does not.* The eyes-closed extension makes exactly that tangible, safely, in thirty seconds.

5. Lesson flow

The in-app wizard has the steps below. **Before each measurement the app asks for a written prediction** in the on-screen journal and quotes it back in the conclusion; the journal is saved to disk by itself.

1. **What are roll and pitch? (3 min).** The app's first step. Show it with your own palm: tilting sideways is *roll*, tilting forward and back is *pitch* — the words pilots use. The button is **Got it**.
2. **The written prediction (2 min).** The second step: *is your desk level? What about the window sill?* And then the good question: *how long do you think you can hold the box perfectly still: one second, ten, a minute?* Almost every child overestimates. The button is **My prediction is written**.

Technical note — two names, two numbers and a circle that turns green

The sensor. MPU-6500. **The box is mounted upside down and the firmware already corrects it** — a flat box gives $AZ \approx +1$ g. **Roll and pitch do not depend on the sign of** AZ, because it enters the formulas squared. From firmware 2.2.5 the readings are smoothed inside the chip itself, at about 20 Hz, so the figures tremble less than on boxes with older firmware.
How it starts. The activity opens from the **Activities** page, from the **Flying Straight** card, with the **Start** button. The window that opens carries the same name.
The two cards. Roll and **Pitch**, both in degrees, with a sign: negative one way, positive the other. Beside them is the **artificial horizon**, with pitch reference marks every ten degrees.
The digital level. The app first asks for the box to be **set down on a flat surface**, hands off, and the **Reading 1** button; then for **the reversal test**: the same box, turned round by 180° on the spot, and the **Reading 2 (turned round)** button. The last two readings stay written under the buttons, with the roll and pitch of each. Until the first reading, the green button stays pale, with a tip underneath; you can still move on.
The "Table or box?" step. Straight after the level, the app does the reversal-test arithmetic itself from the last two readings: it shows them and says whether **the surface is tilted** (the numbers changed sign, and by how much), whether it is **the box's own offset** (the numbers stayed the same), or whether both are close to zero. Above it sits the Junior uncertainty tip: *measure twice in the same place; if the two numbers are close, the reading is good.* Without two readings, the step asks you to go back to the level. See the subsection on the reversal test.
The challenge, exactly. The circle turns green when **both |roll| and |pitch| are below**

5°. It has to be held green for **ten seconds**. **At the first departure from tolerance the timer resets to zero** — it does not resume where it left off. This is the rule that makes the challenge hard, and it is worth stating in advance so that failure does not look like a fault.

The button cannot be pressed before success. At the challenge step the **I did it!** button becomes active only once the circle has stayed green for ten seconds (until then, hovering over it shows *Keep the circle green for 10 seconds first*). A child can no longer move on straight after the message *You tilted! The timer has reset.*

Counting the tries. Under the progress bar, **Tries** goes up by one *every time the circle turns green*. The app's number can therefore be larger than the child's own count, which usually counts only the lifts of the box; the conclusion quotes the app's number.

The cable matters. A taut USB cable pulls the box and tilts it by a few degrees, steadily. Leave it slack on the table or work over WiFi — otherwise a perfectly level surface will look crooked.

The buttons. The green button changes its label at every step: **Got it**, **My prediction is written**, **I tested the surfaces**, **Continue**, **I did it!** and, at the end, **Finish**. **Back** goes one step back; a challenge won stays won.

A box that shows −1 g on AZ while lying flat. If the app flagged **Old firmware**, reflash the box. If the firmware is current, the box has its sensor fitted the other way up: under **Settings → Box names** tick **Motion sensor upside down** for it. With firmware 2.3.0 or newer the tick is kept in the box itself, so it counts on every computer; the box must be connected when you tick it. The activity has no correction of its own.

The journal and the export. What the children write in the journal is saved in the `logs` folder, in a `_journal.txt` file named after the activity, the box and the date. **For this activity exporting the data is rarely needed**: the result is the number of attempts, recorded on the sheet.

3. **The digital level — the tour of surfaces (3 min).** The box is placed in turn on the marked surfaces and roll and pitch are recorded. *Deliberately* put a book under one edge of one of them, so that there is a case you know to be crooked. On the surface that looks most tilted, do **the reversal test**: **Reading 1**, then the box turned round by 180° in the same place and **Reading 2 (turned round)**. The button is **I tested the surfaces**.

4. **Table or box? (1 min).** The app gives the verdict of the reversal test. The question for the class: *which surface in this room is the most level — and is it the table or the box?* The button is **Continue**.

5. **The challenge (4 min).** Every child in turn starts with the box on the desk, lifts it in both hands and tries to hold the circle green for ten seconds. **Count the attempts**; do not time who is fastest. The **I did it!** button becomes active after success.

6. **Why pilots need instruments (2 min).** The final step. Ask the question that ties it together: *if a pilot can see nothing through the window, how do they know they are flying level?* Close with **Finish**.

Debrief: the score is not the point

Do not ask "who did it first". Ask **"on which attempt did you succeed, and what did you change between them?"**

That is the only question that turns the game into a lesson, and it has real answers that children find for themselves: *I held the box in both hands*; *I rested my elbows on the table*; *I started from where the box sat level on the desk*; *I watched the circle, not my hands*; *I kept still and breathed more slowly.* Every one of them is a **deliberate intervention followed by a check** — that is, scientific method, at eight.

Three debrief questions, in order:

- *What stopped you the first time?* — almost always the shaking hand, elbows in the air, or the cable.
- *What did you change, and did it work?* — this is where the intervention gets named.
- *When you turned the box round on the window sill, what happened to the number?* — this tests whether they have grasped that a tilted surface changes sign when the box is turned round, and the box does not.

The score is not the objective. A child who succeeded on the sixth attempt and can say what they changed each time has learned more than one who hit it first time. Say so explicitly to the class *before* the challenge — otherwise the number of attempts becomes an embarrassment instead of a result.

The investigation, in one table

The question. Which surface in our room is the most level?
The hypothesis. Written on the sheet **before** any measuring: rank the four surfaces by how level you think they are. Not changed afterwards.
What I change. One thing only: the **surface** the box stands on.
What I measure. Roll and pitch, in degrees, with their signs.
What I keep the same. The same box, **the same orientation** — with the same edge towards me every time, because otherwise roll and pitch swap over — **the same place on the surface** (the middle, not the edge), and the cable left slack.
How many repeats. Two placings on each surface, lifting the box in between. If the two readings differ a lot, the box was not put in the same place.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken with the box set down, hands off the table and the cable slack. An artefact is anything else: **a hand leaning on the table presses it out of level**, a taut cable pulls the box steadily to one side, and a box on the edge of a desk reads differently from the same desk at the middle. **The decisive test is the** 180° **reversal**: if the number changes sign, the surface is tilted; if it stays the same, it is the box's own offset.

6. Analysis after the activity

6.1 What the data should look like

Situation	Typical value	What it means
A good desk, box in the middle	0–1°	Level. What remains is mostly the box's offset.
The same desk, box turned round by 180°	the same number, the same sign	**The box's offset**, not the tilt of the desk.
The same desk, box on the edge	1–3°	The top sags a little; choose the middle.
Window sill	1–5°	**Often tilted on purpose**, so that water runs off. Turned round, the sign changes.
Floor	0–2°	Usually the most level surface in the room.
With a book under one edge	5–15°	The control case: we know it is crooked.
Box held in both hands, elbows on the table	1–4°, trembling	The challenge can be won.
Box held in one hand, arm in the air	2–8°, trembling	**Which is why both hands and elbows down.**
Box on a table with somebody leaning on it	shifts by 1–4°	The table tilts under the weight.
Taut USB cable	a steady offset of 1–3°	It pulls the box always the same way.

6.2 Questions to ask afterwards

1. You thought your desk was level. What did the instrument say? Who was right — your eyes or the box?
2. Why is it easier with both hands and your elbows on the table? What does your hand do even when you think it is still?
3. The window sill came out tilted. **Is that a builder's mistake?** What might a slight slope there be for?
4. When you turned the box round, one number changed sign and another did not. Which one was the table's fault, and which the box's?
5. A pilot entering cloud can no longer see the horizon. Why can they not rely on what they feel, and what do they use instead?

6.3 Assessment ideas

- **Low stakes:** the child ranks the surfaces from most level to most tilted, using only the colour of the circle or the size of the numbers, without writing them down.
- **Standard:** the child records roll and pitch for each surface, compares them with the prediction, and says which surface surprised them.
- **Extension:** the child explains in writing why they only succeeded at the challenge after several attempts and what exactly they changed — that is, describes an intervention and its result; or says, after the reversal test, whether the tilt came from the surface or from the box.

Common pitfalls and misconceptions

The box held in one hand, arm in the air. The tremor is larger than five degrees. The challenge becomes impossible and the child thinks it is their fault. Say it first: **both hands, elbows on the table**.

Hands on the table during the digital level. A table pressed at one corner tilts by a few degrees. While the box is measuring a surface, everybody takes their hands off the table.

A taut cable. It pulls the box steadily and makes a level surface look crooked. The cable lies slack, or you work over WiFi.

Rotating the box by 90° between measurements. Roll and pitch swap over, ruining the comparison. **The same edge towards you** every time — the only rotation allowed is the deliberate 180° turn of the reversal test.

Measuring on the edge of the desk. Desk tops sag towards the edges. Measure at the middle, always in the same place.

Expecting zero to mean exactly zero. Every box has a small offset, and the last decimals always tremble. The reversal test shows how big the offset is; the five-degree tolerance exists precisely because perfect zero cannot be reached.

Turning the challenge into a speed contest. The number of attempts is a result, not an embarrassment, and whoever succeeds first time has learned nothing. See the debrief box.

Safety and classroom management

The box is held in both hands, over the table. It is small and slippery; for the challenge the child sits with elbows on the table and does not carry it about the room. The box is never thrown from one child to another: it is put down on the table and picked up from there.

Nobody climbs on chairs or desks to reach a high surface — the top of a cupboard is measured only if it can be reached standing on the floor, and otherwise the teacher measures it.

Window sills. Children do not climb on sills and do not lean out of windows to place the box. If the sill is high, the teacher makes the measurement and the children read the screen.

Cables. A USB cable stretched from the laptop to a distant surface is a trip hazard and spoils the measurement besides. The cable stays slack; WiFi solves both problems.

Books placed under edges are arranged so that nothing can slide off and fall on feet.

Adaptations for pupils with special educational needs

Colour only, no numbers. This is the main adaptation and it is not a concession: the child plays and ranks the surfaces by watching *the colour of the circle* and how quickly it turns green, never reading roll and pitch. The objective — that surfaces differ and that an instrument can say so — is met in full with no signed numbers at all.

The challenge, with support. For children with tremor or difficult fine control, the challenge is done with the box **left set down** on the desk, and their task is to *not touch the table* for ten seconds and to announce when the circle turns green. It is a real task and it can be won.

Three real roles for children with motor difficulties. The roles rotate at every surface:

- **the hands keeper** — checks before each attempt that nobody is touching the table and that the cable is slack, and **has the right to void an attempt**; this is the most valuable role in the group, because they decide which attempt is fair;
- **the screen reader** — says the two numbers, or just the colour;
- **the timekeeper** — counts the ten seconds aloud.

Frustration. A resetting timer can be hard to bear. **Announce the rule in advance** and say openly that most adults need several attempts — which is true. A child who knows failure is expected takes it entirely differently.

If things do not go to plan

Nobody can do the challenge. Check, in order: are the children holding the box in both hands, elbows on the table? do they start from a level surface — one that reads under 1° when nobody touches it? is the cable slack? If it still does not work, switch one group to the left-on-the-desk version from the adaptations box.
Everybody succeeds first time. The group is calm and the hands are steady — move to the eyes-closed extension, which is far harder and far more interesting.
The numbers never come near zero. If they stay at the same value wherever you put the box, check that there is nothing under it and do the reversal test on the floor: **a number that does not change sign when the box is turned round is the box's offset**. Under one degree is normal; several degrees means a box to be checked.
You have one box and thirty children. Screen projected, the challenge in turn, and **the roles from the adaptations box for everybody else** — the hands keeper, the screen reader and the timekeeper change at every attempt, so nobody is waiting idle.
The box will not connect. Do not stop the lesson. The tour of surfaces can be done with **a spirit level**, if you have one — and the reversal test works just as well with it —, and the discussion about the tilted sill and about pilots in cloud is unchanged. Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **NASA** — `https://www.nasa.gov` — search their pupil material on flight and cockpit instruments; suitable for the activity's final step.
- **Institute of Physics** — `https://www.iop.org` — search their primary material on forces and balance, written for teachers who are not specialists.
- **STEM Learning** — `https://www.stem.org.uk` — search for flight and balance for ages 7–11; many lesson plans this measurement slots straight into.
- **Royal Aeronautical Society** — `https://www.aerosociety.com` — material on the principles of flight, if the activity provokes questions about aircraft.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the forces and balance simulations.
- **Skybrary** — `https://skybrary.aero` — for the teacher, on spatial disorientation in flight; it explains why the instrument beats the sensation.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *what did you change between attempts?* Collected in the last two minutes, it says more about the class's understanding than any mark — and, in this activity, more than the number of attempts.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Junior level the three axes look like this:

- **Measurement practice** — the child sets the box on the surface, places it with the same edge towards them every time, takes their hands off the table and leaves the cable slack.
- **Data interpretation** — the child says which surface is more level and by how much, and compares their ranking with the prediction.
- **Scientific reasoning** — the child describes an intervention — what they changed between attempts — and explains why it worked.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the children fill in is a separate document: `docs/en/student_sheets/flying_straight.pdf`. It has two pages — the prediction, the ranking of surfaces and their table on the first; the reversal test, the challenge, what I changed between attempts, and the conclusion after them — so it is printed double-sided, one sheet per child.

Three things to know before printing it. **The sheet asks for the ranking *before* measuring**, as a prediction, and leaves room beside it for the real ranking — the collision between them is the lesson, and it has to be visible on the same page. **The challenge table asks for "what I changed", not just the number of attempts**: without that column the game stays a game. **The number column has a "green / not" column beside it**, precisely for the child who does not work with signed numbers: they circle the second and leave the first empty. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.